

ME416

Ground Vehicle Dynamics

Tires, Rolling Resistance and More

Fall 2018

[1]

Wheel Resistance

- The wheel resistance, F_w consists of

$$F_w = F_{w,roll} + F_{w,rd} + F_{w,sl}$$

- Rolling resistance, $F_{w,roll}$, due to
 - Flexing resistance
 - Friction resistance
 - Fan resistance
- Road resistance, $F_{w,rd}$, due to
 - Damping resistance on an uneven road
 - Flush resistance on wet road
 - Deformation resistance on a malleable road
- Slip resistance, $F_{w,sl}$, due to cornering
- Resistance due to wheel bearings etc. although this is generally considered to be negligible

Fall 2018

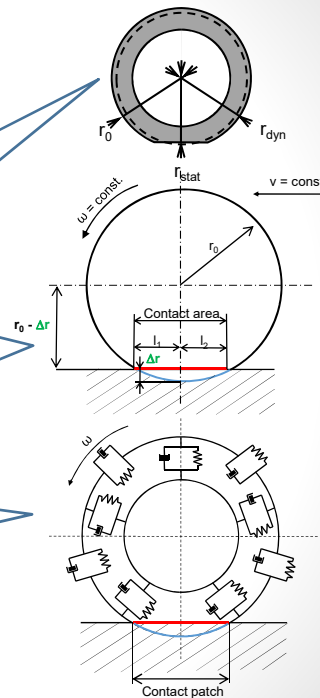
[2]

Rolling Resistance

- Flexing resistance results from the loss of energy due to the internal friction of the tire where
 - r_0 is the unloaded radius
 - r_{stat} is the statically loaded $r_{stat} < r_0$
 - r_{dyn} is the dynamic (rolling) radius $r_{stat} < r_{dyn} < r_0$

- With every revolution each point of the tire will pass through the contact patch
 - The original length of the tire (blue) is shortened to the chord length (red) of the contact patch
 - The reduces the radius from r_0 to r_{dyn} and compresses the tire by Δr

- The tire behaves as a spring-damper units
 - Connected in series about the circumference
 - Connected in parallel radially



Fall 2018

3

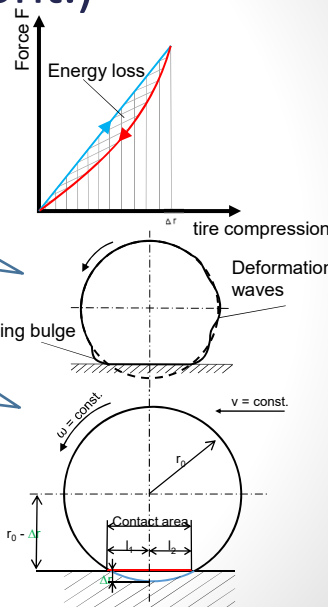
Rolling Resistance (Cont.)

- Entering the contact patch the tire gets compressed, radially and tangentially, pre-tensioning the system
- Exiting the contact patch the tire compression is released
 - Due to damping the force within the rubber is reduced and energy absorbed remains as heat in the tire

- At a certain velocity the flexing resistance rises greatly
- The excitation is much higher than the damping of the tire and deformation waves and a rolling bulge will form
- This behavior defines the tires high speed limit as it will heat absorption rises and quickly destroys the tire

- With every rotation the the the contact area actually
 - Gets contracted in l_1
 - Then extended in l_2
 - Resulting in friction between the tire and the road

- The airflow around the tire also induces losses and is called the fan resistance

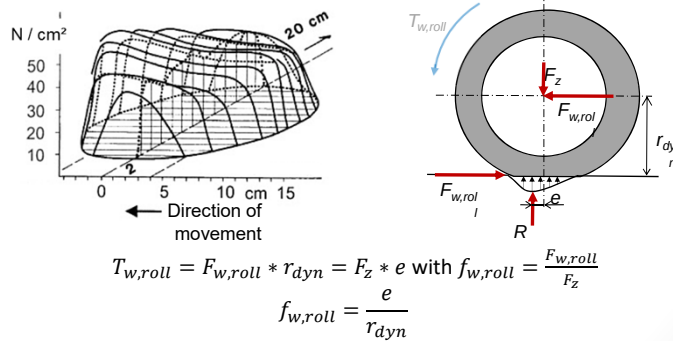


Fall 2018

4

Rolling Resistance (Cont.)

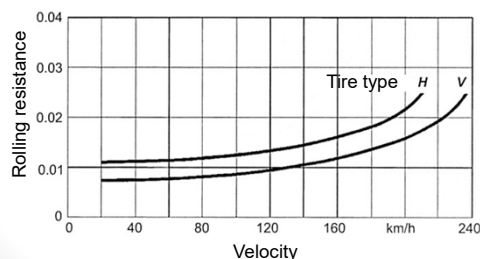
- At 140kph (87mph) the rolling resistance for a road car is approximately broken down as
 - Flexing resistance: 77-96.5%
 - Friction resistance: 2-20%
 - Fan resistance: 1.5-3%
- We observe the pressure within the contact patch and the resulting assymetric load distribution to develop a relationship between eccentricity and the rolling resistance coefficient



[5]

Rolling Resistance (Cont.)

- Rolling resistance is considered constant
 - For different wheel load (if pressures adjusted) within a limited load range
 - For increasing velocity as long as the velocity is well below the speed limit (rating) of the tire
- Rolling resistance is also, in general, reduced as wheel diameter increases
 - For a given contact area (length) the compression of the wheel is less for larger diameter tire reducing the flexing and friction losses

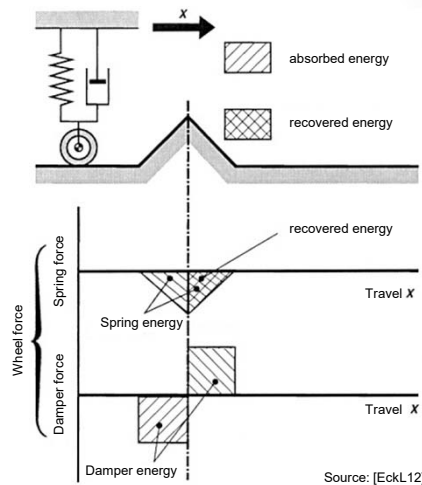


Tire types:
 H up to 130mph (210 kph)
 V up to 150mph (240 kph)

[6]

Road Drag, Uneven Road

- The resistance due to the tire interaction with an uneven road is a function of its spring/damper behavior
 - The energy stored in the tire spring is recovered as the tire passes the obstacle
 - The energy absorbed by the tire damper is lost
 - The spring/damper characteristics will be a function of tire construction and pressure

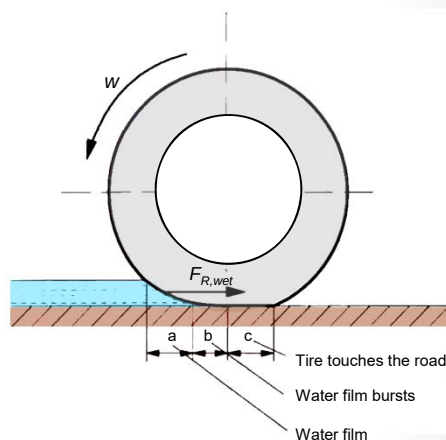


Fall 2018

[7]

Road Drag, Wet Road

- The flush resistance results from the displacement of water in the contact area of the tire on a wet road
- It depends on
 - Thickness of the water film
 - Velocity
 - Section width of the tire
 - Tread pattern
 - Tread depth
- The length of section 'c' where the tire touches the road gets smaller as
 - The height of the water film increases
 - The velocity increases
 - The section width increases
 - The tread depth reduces
- Ultimately, the tire aquaplanes when section 'c' vanishes and the tire is essentially unable to generate any lateral or longitudinal forces

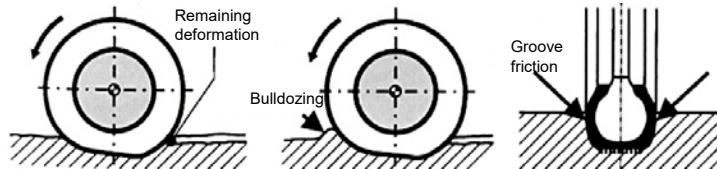


Fall 2018

[8]

Road Drag, Malleable Road

- Unlike a solid road surface like tarmac or concrete a malleable road deforms due to a passing vehicle (or tire)
- The deformation consists of 3 components



- The deformation and therefore resistance can be reduced by lowering the air pressure, increasing the contact patch area

Road	f_w
Solid ground like tarmac or concrete	0,01 – 0,015
Gravel road	0,02 – 0,04
Dirt road	0,04 – 0,15
Sand	0,15 – 0,4

Fall 2018

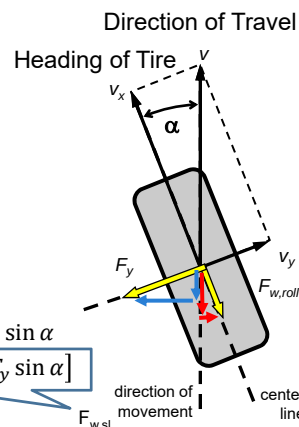
[9]

Slip Angle Resistance

- Recall that a slip angle is required to generate lateral force
- Also recall that the lateral force and longitudinal forces are along the tire coordinate axis defined by the heading of the tire
- For solid ground this results in

$$\begin{aligned}
 F_w &= F_{w,roll} \cos \alpha + F_y \sin \alpha \\
 &= F_{w,roll} - F_{w,roll} * F_{w,roll} * \cos \alpha + F_y \sin \alpha \\
 &= F_{w,roll} + [-F_{w,roll} * (1 - \cos \alpha) + F_y \sin \alpha]
 \end{aligned}$$

$$\begin{aligned}
 F_{w,sl} &= F_y \sin \alpha - F_{w,roll} * (1 - \cos \alpha) \\
 f_{w,sl} &= \frac{F_y \sin \alpha - F_{w,roll} * (1 - \cos \alpha)}{F_z} \\
 &= \frac{F_y \sin \alpha}{F_z} - f_{w,roll} * (1 - \cos \alpha)
 \end{aligned}$$



Fall 2018

[10]

Slip Angle Resistance (Cont.)

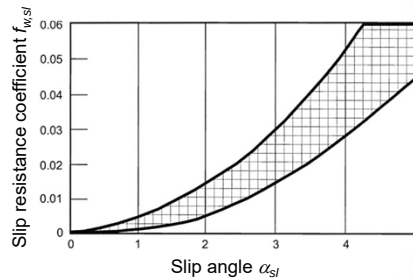
- For small slip angles the lateral force is proportional to the angle

$$F_y = C_\alpha \propto$$

$$f_{w,sl} = \frac{F_y \sin \alpha}{F_z} - f_{w,roll} * (1 - \cos \alpha)$$

where $C_\alpha \propto \alpha^2$

where
 $\sin \alpha \approx \alpha$ and $\cos \alpha \approx 1$
 therefore, the slip angle resistance (or induced drag) increases by the square of the angle



Fall 2018

[11]

Overall Rolling Resistance (Straight Ahead Travel)

Wheel-road combination		f_w
Passenger car wheel on	concrete, asphalt	0,01
	small cobblestone	0,013
	big cobblestone	0,013
	gravel	0,02
	dirt track	0,05
	farmland, soil	0,1 – 0,35
Commercial car wheel on	concrete, tarmac	0,006 – 0,01
Tractor wheel on	farmland, soil	0,14 – 0,24
Caterpillar track on	farmland, soil	0,07 – 0,12
Railroad wheel on	rail	0,001 – 0,002

Fall 2018

[12]

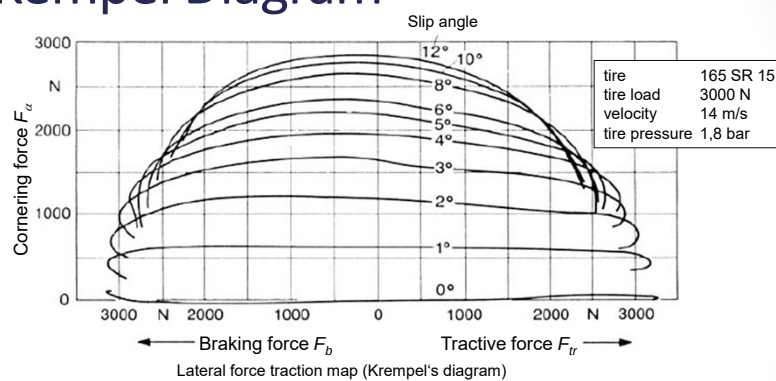
Combined Operation

- Except for very rare occasions (or very, very poor drivers) there will never be a prolonged instance of only cornering, traction or braking forces acting upon the tires
- Therefore, we must have a method of evaluating the performance characteristics of the tire under combined operation
- Facilities for evaluating tires under combined operation are limited as is the data

Fall 2018

[13]

Combined Operation – Kempel Diagram



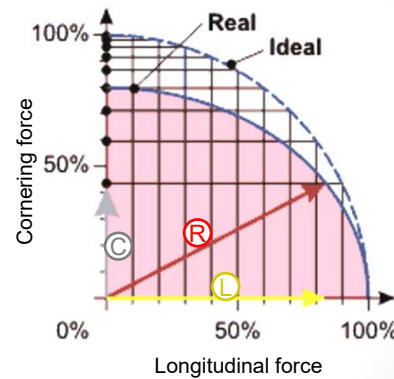
Fall 2018

[14]

- Note the shape of a semi-circle or ellipse (if mirrored)
 - This is different from the common concept of the friction circle (Kamm circle) which is an idealization of the grip available

Combined Operation – Kamm

- In reality the transferable longitudinal forces are greater than the lateral forces
- This results in the effect that near maximum lateral force (cornering), small reductions in lateral force can allow a large change in available transferable longitudinal force
- Why?
 - Tire spins in the longitudinal direction, not lateral
 - Effects of the tread pattern bias

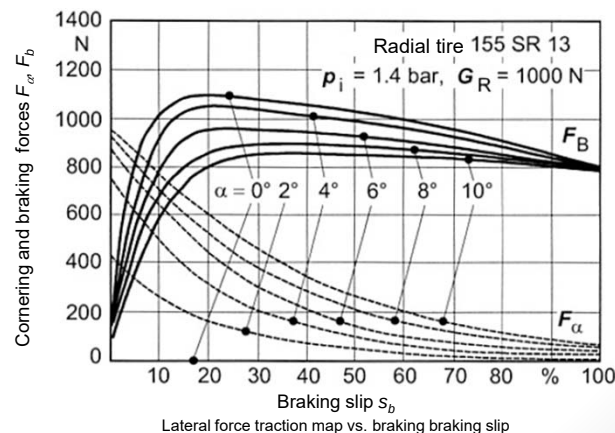


Fall 2018

[15]

Combined Operation

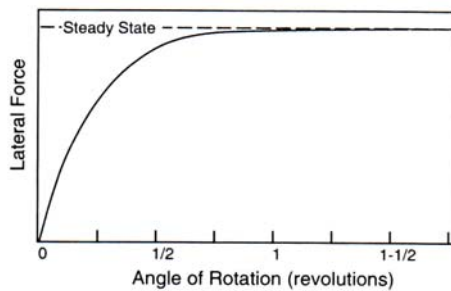
- Note
 - The higher the slip angle the lower the transferable braking forces
 - At maximum of braking forces there can still be significant lateral forces transferred



Fall 2018

[16]

Tire Response



- A tire's response to a lateral load or steer input is not instantaneous
- To generate lateral force the lateral deformation of the tire must occur
- This deformation time is closely related to the rotation of the tire
 - Typically, it takes one-half to one full revolution of the tire to establish
 - This distance is often referred to as the Relaxation Length

Fall 2018

[17]

Tire Response (2)

- The relaxation effect is the primary reason for the reduction in lateral force over rough surfaces
- Each time the vertical or lateral load changes the deformation must be established/reestablished, i.e.
 - When the load diminishes, slip will occur over the entire length of the contact patch and the deformed area will relax
 - When the load returns the tire must roll through its relaxation length to reestablish the contact patch deformation
- This is why for racing purposes the primary goal of the suspension is to minimize the variation of load on the tire
 - The maximum lateral force can be reduced up to 60% due to varying loading!

Fall 2018

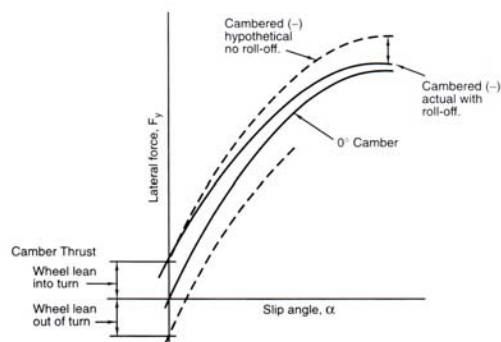
[18]

Backup Slides

Fall 2018

[19]

Camber Effects – Lateral Force

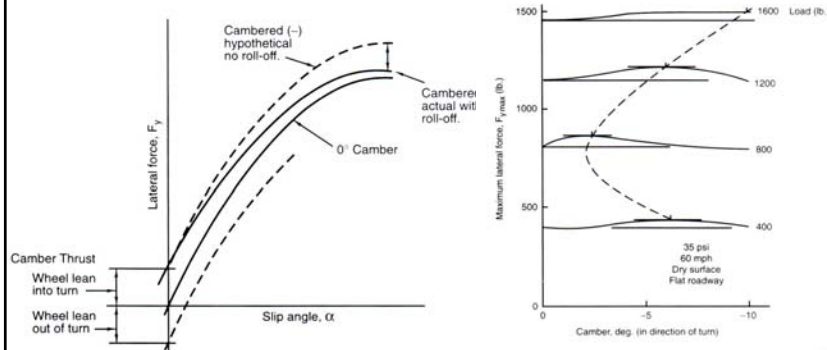


- A slight increase in maximum lateral load is observed with camber
- The exact mechanism is not understood
- A typical result of camber effect on lateral load is shown to the top right
- The maximum lateral load vs. camber and normal load for a racing tire is shown lower right
- Additional data are presented in the following slide

Fall 2018

[20]

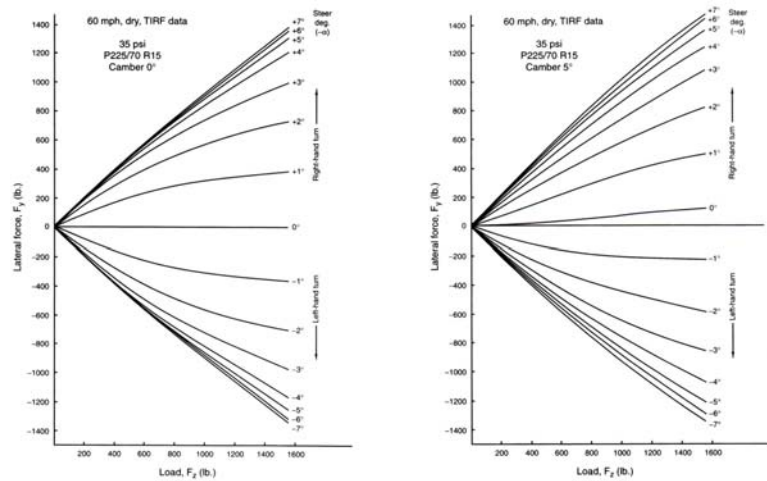
Camber Effects – Lateral Force (2)



Fall 2018

(21)

Camber Effects – Lateral Force (2)



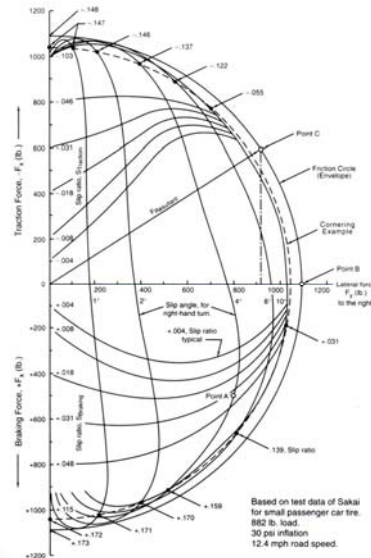
Lateral Force at 0° Camber

Lateral Force at 5° Camber

Fall 2018

(22)

Other Tire Effects

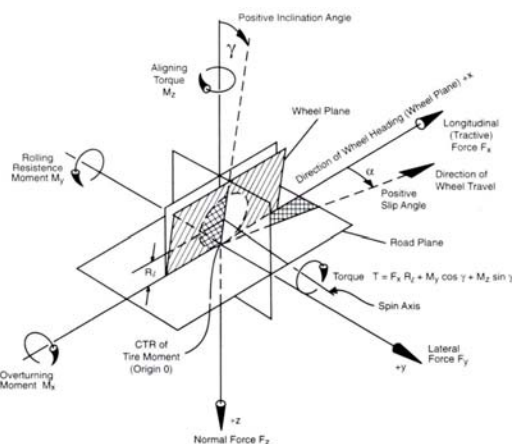


- Reference Milliken and Gillespie for a general discussion of these effects
 - Tire Pressure
 - Tire Temperature
 - Speed Effect
 - Conicity and Ply Steer
- A friction circle or ellipse may be generated for a tire just as for a complete vehicle
 - A friction circle generated from the Sakai data is presented at right
 - Complete tire data like this is very rare
 - Reference Milliken for more detail
 - Additional discussion in Gillespie and lannce

Fall 2018

[23]

Tire Forces and Moments

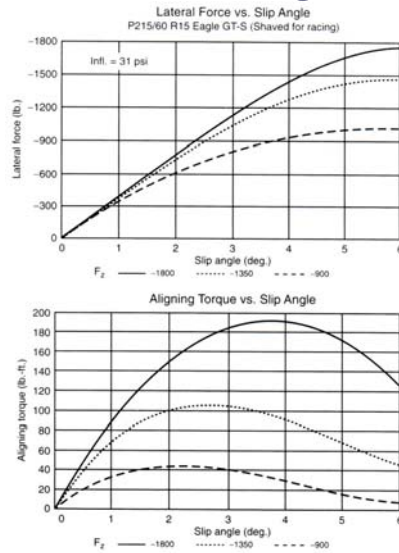


- The SAE convention is shown to the right
- As a designer you will be interested in determining
 - Normal force
 - Tractive force
 - Lateral force and associated induced drag
 - Aligning torque (moment)
 - Overturning moment
 - Rolling resistance
 - Torque about the wheel spin axis
- Note that due to the suspension geometry and tire deformation the determination of these forces and moments are not as trivial as you might assume
- Reference Milliken, Gillespie and lannce

Fall 2018

[24]

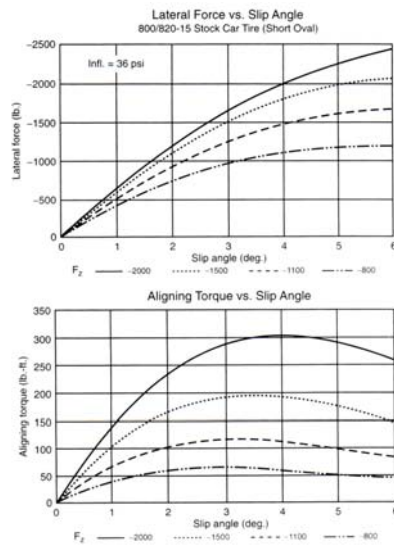
Goodyear Data – Eagle GT-S



Fall 2018

(25)

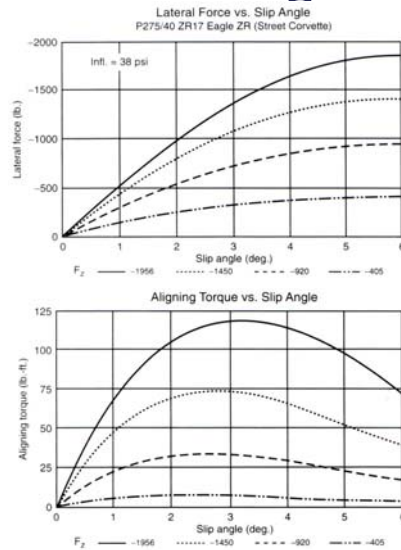
Goodyear Data – Stock Car Tire



Fall 2018

(26)

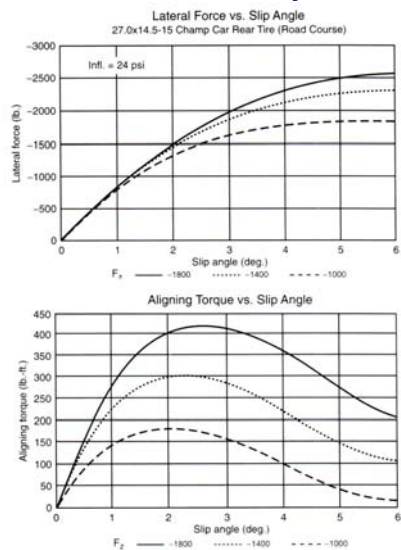
Goodyear Data – Eagle ZR



Fall 2018

[27]

Goodyear Data – Indy Car RC Rear



Fall 2018

[28]

Goodyear Data – F1 Front

